

Image Operations

Negative ($s = L-1-r$): Inverts all intensities. Dark becomes bright, bright becomes dark.

Log Transform ($s = c \log(1+r)$): Compresses high intensities, expands low ones. Reveals detail in dark regions.

Power-Law / Gamma ($s = cr^\gamma$): $\gamma < 1$: brightens dark regions. $\gamma > 1$: darkens image, suppresses low intensities.

Contrast Stretching: Maps a narrow intensity range to the full $[0, L-1]$ range. Increases overall contrast.

Intensity-Level Slicing: Highlights a specific intensity band while suppressing or preserving all others.

Bit-Plane Slicing: Decomposes into 8 binary planes. Higher planes carry structure; lower planes carry fine detail/noise.

Thresholding ($s = 0$ or $L-1$): Produces binary image. Separates foreground from background.

Histogram Equalization: Flattens histogram to use full intensity range. Increases global contrast.

Histogram Matching: Reshapes histogram to match a specified target distribution.

Local Histogram Processing: Applies equalization in sliding neighborhoods. Reveals local detail that global methods miss.

Addition / Averaging: $\bar{g}(x, y) = \frac{1}{K} \sum g_i(x, y)$. Reduces noise variance by $1/K$. Images must be registered.

Subtraction: Highlights differences between two images. Removes static background.

Multiplication / Division: Mask with ROI to isolate regions. Division corrects uneven illumination/shading.

Smoothing (Box / Gaussian): Averages neighborhood pixels. Reduces noise but blurs edges.

Median Filter (nonlinear): Replaces pixel with neighborhood median. Removes salt-and-pepper noise while preserving edges.

Sharpening (Laplacian, Unsharp Mask): Enhances edges and fine detail. Makes transitions crisper.

NOT: swaps foreground/background. **AND**: intersection of two masks. **OR**: union of two masks. **XOR**: pixels in one mask but not both.

Image Overlap

$$\cup = \text{OR} \quad \cap = \text{AND} \quad A^c = \text{NOT}$$

Filter Kernels

Smoothing (Low-Pass)		
Box	Weighted Avg	Gaussian
$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$	$h(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$

Sharpening (High-Pass)		
Laplacian	Laplacian (diag)	Unsharp ($k=1$)
$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	$\frac{1}{9} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$

Edge Detection (Gradient)		
Sobel G_x	Sobel G_y	Prewitt G_x
$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$

Fourier Transform

Forward (spatial $\rightarrow$ frequency):

$$F(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt$$

Inverse (frequency $\rightarrow$ spatial):

$$f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu$$

Constant h over $[a, b]$:

$$\int_a^b h e^{-j2\pi ut} dt = \frac{h}{j2\pi u} (e^{-j2\pi ua} - e^{-j2\pi ub})$$

Linear vs Nonlinear

Linear: output is weighted sum of inputs. Obeys superposition: $T[af_1 + bf_2] = aT[f_1] + bT[f_2]$. Examples: box filter, Gaussian, Laplacian.

Nonlinear: does not obey superposition. Examples: median filter, min/max filter.

Noise Types

Gaussian: random intensity variations across all pixels. Follows normal distribution. Reduced by averaging/Gaussian filtering.

Salt & Pepper: random pure white (255) and pure black (0) pixels. Best removed by median filter, not averaging.

Histogram Matching Method

Goal: transform $p_r(r) \rightarrow p_z(z)$ via uniform bridge s .

$$r \xrightarrow{T_r} s \text{ (uniform)} \xrightarrow{T_z^{-1}} z \quad \text{so} \quad z = T_z^{-1}(T_r(r))$$

Step 1: Equalize input. $s = T_r(r) = (L-1) \int_0^r p_r(w) dw$

Step 2: Equalize desired output. $s = T_z(z) = (L-1) \int_0^z p_z(w) dw$

Step 3: Set $T_r(r) = T_z(z)$, solve for z in terms of r .

Example: $p_r(r) = \frac{4r}{3(L-1)^2}$, want $p_z(z) = \frac{2z}{(L-1)^2}$

$$T_r(r) = (L-1) \int_0^r \frac{4w}{3(L-1)^2} dw = \frac{2r^2}{3(L-1)}$$

$$T_z(z) = (L-1) \int_0^z \frac{2w}{(L-1)^2} dw = \frac{z^2}{L-1}$$

$$\text{Set equal: } \frac{2r^2}{3(L-1)} = \frac{z^2}{L-1} \Rightarrow z^2 = \frac{2}{3}r^2 \Rightarrow z = \sqrt{\frac{2}{3}} r \approx 0.816 r$$